Stastical Quality Control :-

Quality – Quality means fitness for use where fitness is defined by the customer itself who is using that product. The quality of a product refers to the degree ~~in~~ to which it satisfies customer expectation. Quality has no specific meaning unless it is related to specific function or performance of a product.

Quality Cost –

- Failure cost
  - Internal
  - External
- Appraisal cost
- Preventive cost

Failure Cost – It is the cost associated with producing defective parts within the production system.
Appraisal Cost – It is the cost associated with finding & searching out defective part within the prodn system.
Preventive Cost – It is the cost associated with minimising failure & appraisal cost.

Value of Quality –
The return obtained directly or indirectly by an organisation due to good quality of their product is termed as value of quality. Good quality can earn good response from the customer, firm price policy, increase in market share, higher percentage of successful tender bids & other benifits to the income of organisation.

SQC

→ Implementation of quality at policy, design, mfg & installation stages is essential to ensure that the end product or service meets the required quality standards. Some steps that can be taken to implement quality at each stage are as follows:

① Policy stage:

At this stage, quality can be implemented by setting policies and procedures that promote quality throughout the organization. Following are the steps:

→ Develop a quality policy statement that outlines the company's commitment to quality.

→ Establish quality objectives that align with the company's overall goals.

→ Define procedures for implementing and monitoring quality standards.

→ Assign roles & responsibilities to ensure that quality standards are met.

② Design stage:

→ At this stage, quality can be implemented by ensuring that the design meets the required specifications & standards. Following steps can be taken:-

→ Clearly define design requirements + specifications.

→ Conduct design reviews to ensure that the design meets the required standards.

→ Use appropriate design tools + techniques to ensure that the design is robust + reliable.

→ Conduct testing + validation to ensure that the design meets the required quality standards.

③ Manufacturing stage:

At this stage, quality can be implemented by ensuring that the manufacturing process is controlled + monitored to meet the required quality standards. Following steps can be taken:

→ Develop a mfg plan that includes procedures for controlling + monitoring the process.

→ Implement process controls such as SPC (statistical process control) to monitor process variability + take corrective actions when necessary.

→ Conduct regular audits of the mfg process to ensure that quality standards are being met.

→ Train personnel in the proper use of equipment + procedures to ensure that quality standards are being met.

④ Installation stage:

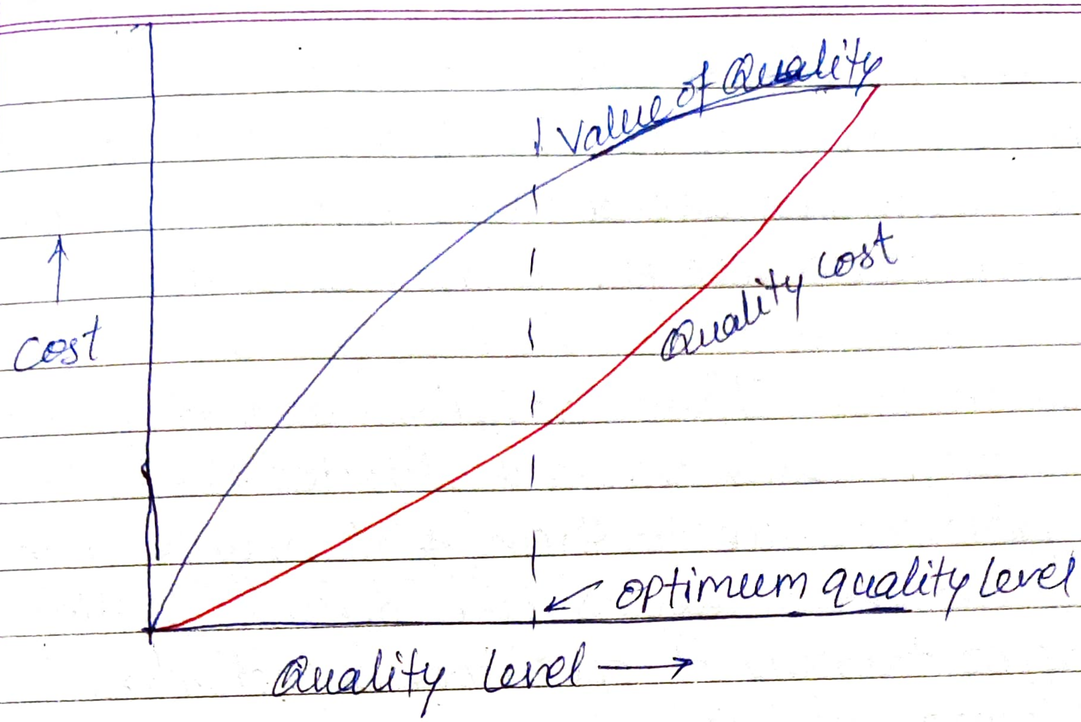
At this stage, quality can be implemented by ensuring that the installation is done correctly + meets the required quality standards. Following steps can be taken:-

→ Develop an installation plan that ~~includes~~ includes procedures for ensuring that the installation is done correctly

→ Train personnel in the proper installation procedures & techniques.

→ Conduct regular inspections of the installation to ensure that quality standards are being met.

→ Document the installation process to ensure that it can be replicated & improved upon in the future.



Quality Control & Inspection-

Inspection simply means checking & shorting out defective product whereas quality control is a broadal term which include no. of activities including inspection & regulate the quality of future production. In quality control if there is defective part we search for the reasons behind defective part & also include the steps to be taken so that it may no be repeated in future.

Variation

- Common Cause or Chance or Random cause
- Assignable Cause

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Common Cause – These are difficult to trace & difficult to control even under the best condition of production & These variations are always within the limit & defective parts are not produced due to them.

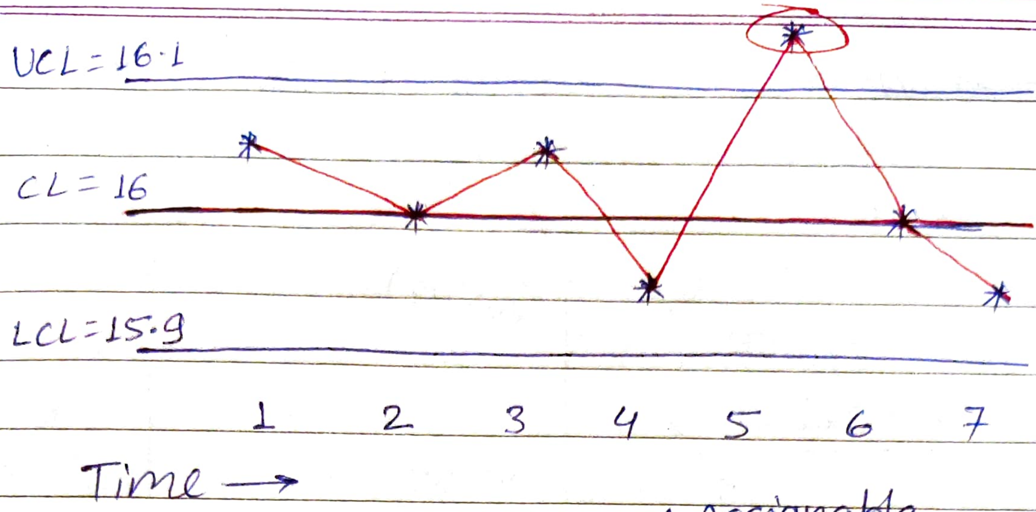
Assignable Cause – These variations are of greater magnitude which can be easily traced & detected. Defective parts are produced due to them & it may be due to some particular reasons like machine setting change, improper training of operator, defective raw material, tool wear, m/c vibrations etc.

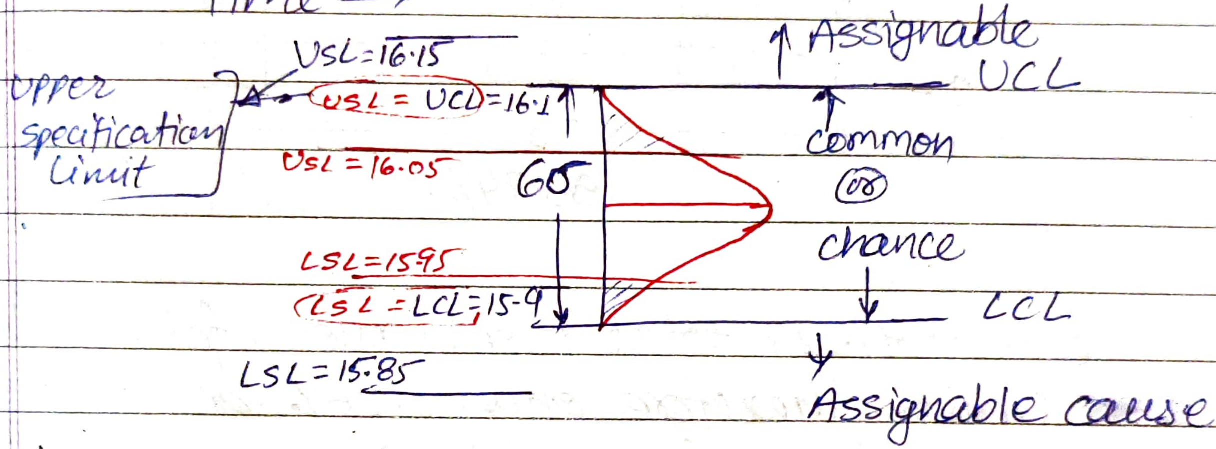
1) Type-I-Error → When there is no problem within the system but still we think that there is some assignable cause of variation.

2) Type-II-Error → When there is some problem within the system but still we think that there is no assignable cause of variation.

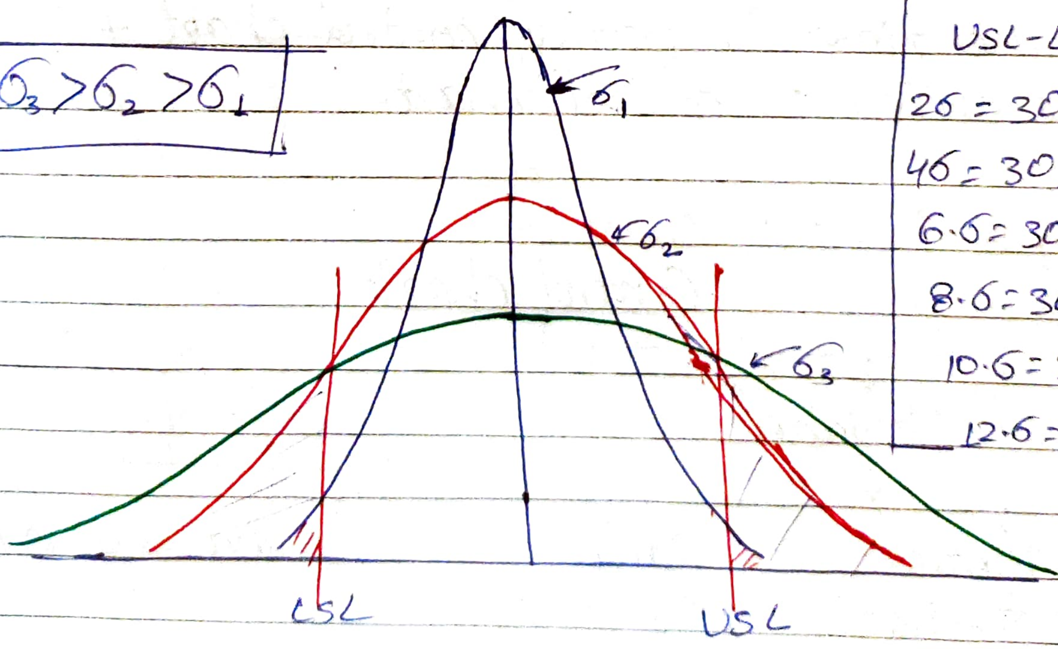
Control Chart :-

Control chart is a graph used to study how a process changes ~~its~~ over time. In these charts observations are plotted in time order. Control chart has a central line for average, an upper line for upper control limit & a lower line for a lower control limit.





$\boxed{\sigma_3 > \sigma_2 > \sigma_1}$



$USL - LSL = 30$

$2\sigma = 30, \ \sigma = 15$

$4\sigma = 30, \ \sigma = 7.5$

$6 \cdot \sigma = 30, \ \sigma = 5.0$

$8 \cdot \sigma = 30, \ \sigma = 3.75$

$10 \cdot \sigma = 30, \ \sigma = 3.0$

$12 \cdot \sigma = 30, \ \sigma = 2.5$

$\boxed{USL - LSL = \uparrow x \cdot \sigma \downarrow}$

↑ Any Number

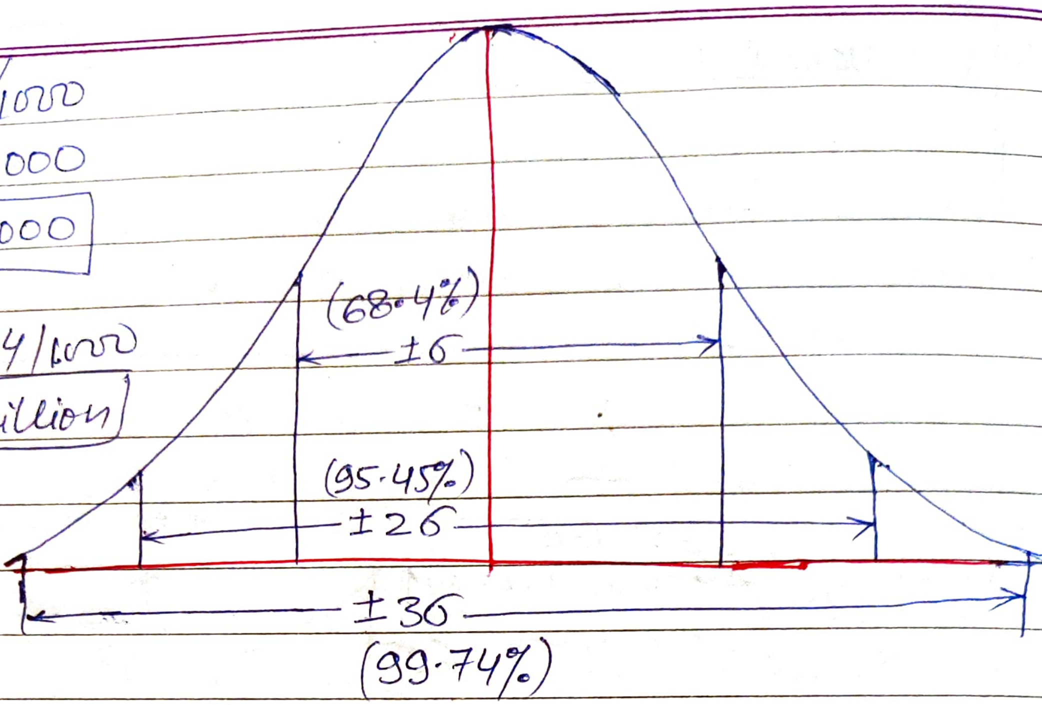
$2\sigma = 320/1000$

$4\sigma = 50/1000$

opt^m costwise $\boxed{6\sigma = 3/1000}$

$8\sigma = 0.184/1000$

Quality wise $\boxed{12\sigma = 3.2/\text{million}}$



Since 99.74% of total observations falls within ±3σ limits therefore only 0.26% defective parts are produced. Therefore ±3σ limits are selected most of the time for plotting control chart & these charts are called 3σ control chart.

Types of control chart –

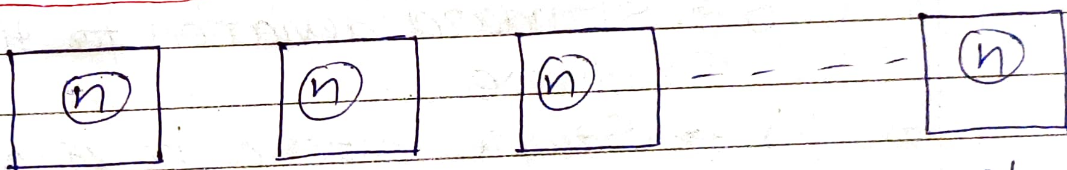
Control charts
- Variable
  - Mean (or) $\bar{X}$-chart
  - Range (or) R-chart
- Attribute
  - P-chart
  - C-chart

Variable – These charts are applied to data that follow a contineous distribution & can be measured on contineous scale. These data are contineous & are assumed to follow normal distribution.

These data can be measured in fractions or decimals like temp., distance, weight, density etc.

Attribute – These data are counted & can not have fraction or decimal. These data arrives while determining either the presence or absence of something like success or failure, good or bad, defective or non-defective etc. These data are discontinuous & are assumed to follow Binomial distribution.

Variable Control Chart



Samples:	1	2	3	----	N
Mean:	$\bar{X}_1$	$\bar{X}_2$	$\bar{X}_3$	----	$\bar{X}_N$
Range:	R_1	R_2	R_3	----	R_N

Grand Averge or Average of sample mean $\bar{\bar{X}} = \dfrac{\bar{X}_1 + \bar{X}_2 + \bar{X}_3 + \cdots + \bar{X}_N}{N}$

Average Range, $\bar{R} = \dfrac{R_1 + R_2 + R_3 + \cdots + R_N}{N}$

1) Mean Chart – It shows the centering of the process or in other words it shows the variation in the average of sample.

2) Range Chart – It moniters the dispersion or variation of process & it is a measure of spread of data (sample

Control Limit

$\bar{x}$ - chart

$$CL = \bar{\bar{x}}$$
$$UCL = \bar{\bar{x}} + 3 \cdot \sigma_{\bar{x}}$$
$$LCL = \bar{\bar{x}} - 3\sigma_{\bar{x}}$$

where, $\boxed{\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}}$

$$\boxed{UCL = \bar{\bar{x}} + 3 \cdot \frac{\sigma}{\sqrt{n}}}$$

$$\boxed{LCL = \bar{\bar{x}} - \frac{3 \cdot \sigma}{\sqrt{n}}}$$

where $\sigma_{\bar{x}}$ → is the standard deviation for the sample mean.

σ → standard deviation for the process or universe

n → Sample size or no. of observation in each sample.

$\bar{R} = \sigma \cdot d_2 \Rightarrow \boxed{\sigma = \frac{\bar{R}}{d_2}}$

$\boxed{UCL = \bar{\bar{x}} + \frac{3 \cdot \bar{R}}{d_2\sqrt{n}}}$ & $\boxed{LCL = \bar{\bar{x}} - \frac{3 \cdot \bar{R}}{d_2\sqrt{n}}}$

$\boxed{UCL = \bar{\bar{x}} + A_2 \cdot \bar{R}}$ & $\boxed{LCL = \bar{\bar{x}} - A_2\bar{R}}$ where $A_2 = \frac{3}{d_2\sqrt{n}}$

Where, d_2 & A_2 are the constant factor whose value depends upon the same size n

R - chart

$$CL = \bar{R} = \sigma \cdot d_2$$
$$UCL = \sigma \cdot d_2 + 3 \cdot \sigma \cdot d_3$$
$$LCL = \sigma \cdot d_2 - 3\sigma \cdot d_3$$

Now as $\bar{R} = \sigma \cdot d_2 \Rightarrow \boxed{\sigma = \frac{\bar{R}}{d_2}}$

$UCL = \bar{R} + 3\frac{\bar{R} \cdot d_3}{d_2}$ & $LCL = \bar{R} - 3\cdot\frac{\bar{R} d_3}{d_2}$

$UCL = \bar{R}\left[1 + 3\cdot\frac{d_3}{d_2}\right]$ & $LCL = \bar{R}\left[1 - 3\frac{d_3}{d_2}\right]$

$\boxed{UCL = D_4 \cdot \bar{R}}$ & $\boxed{LCL = D_3 \cdot \bar{R}}$

where, $D_4 = \left(1 + 3\cdot\frac{d_3}{d_2}\right)$ & $D_3 = \left(1 - \frac{3 d_3}{d_2}\right)$

for $n < 7$, $D_3 = 0$

where, d_2, d_3, D_3 & D_4 are the constant factors whose value depends only upon sample size n.

pb The following data give reading for 10 samples of size 8 each in the production of certain component. Draw the control chart for mean + range and point out which sample if any are out of range.

Samples	1	2	3	4	5	6	7	8	9	10
Mean	5.4	5.1	5.4	4.9	5.2	4.7	5.1	5.0	5.0	5.2
Range	0.4	0.7	0.7	0.8	0.9	0.6	0.5	0.6	0.7	0.6

for $n = 8$, $d_2 = 2.847$, $D_3 = 0.136$, $D_4 = 1868$

for $n = 10$, $d_2 = 3.112$, $D_3 = 0.184$, $D_4 = 2.113$

Sol[4] — $\bar{\bar{X}} = 5.1$ $\quad \bar{R} = 0.65$

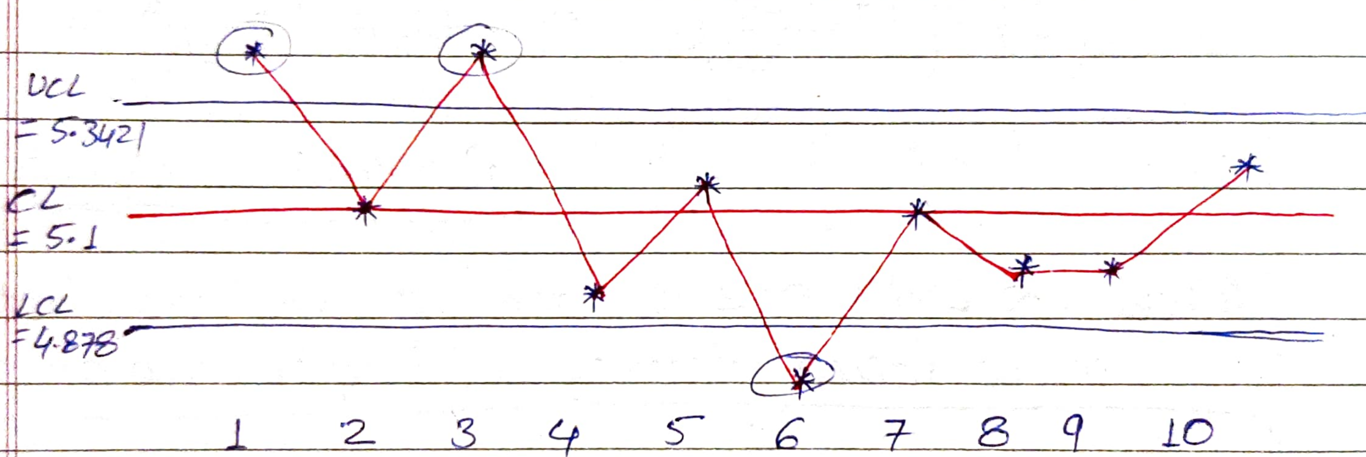
$$\bar{R} = \sigma \cdot d_2 \Rightarrow \sigma = \frac{\bar{R}}{d_2} = \frac{0.65}{2.847} = 0.2283$$

$\bar{X}$-chart

$$CL = \bar{\bar{X}} = 5.1$$

$$UCL = \bar{\bar{X}} + \frac{3 \cdot \sigma}{\sqrt{n}} = \frac{5.1 + 3 \times 0.2283}{\sqrt{8}} = 5.3421$$

$$LCL = \bar{\bar{X}} - \frac{3 \cdot \sigma}{\sqrt{n}} = 4.8578$$



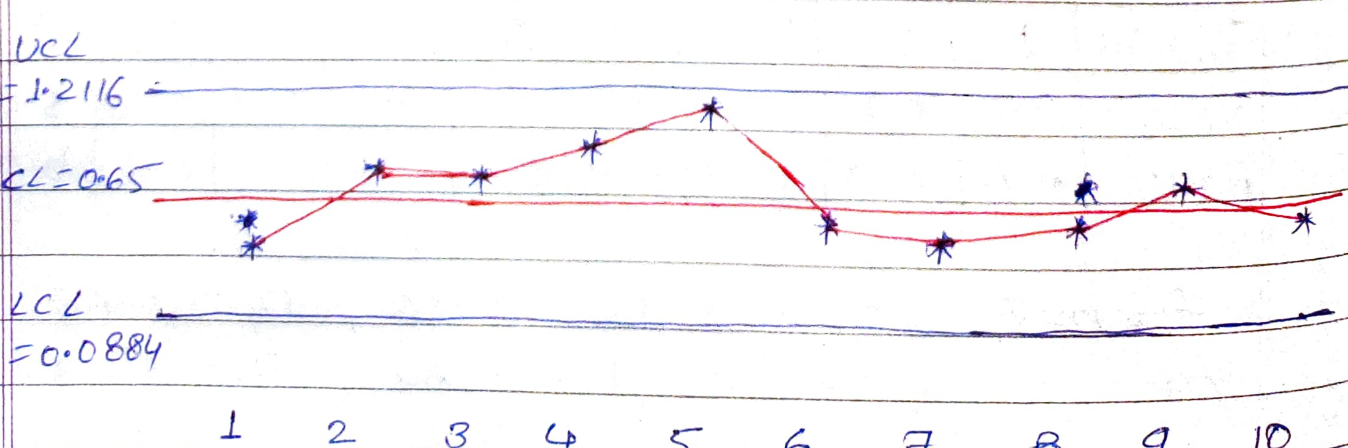
3 → observations are out of control

R-chart

$$CL = \bar{R} = 0.65$$

$$UCL = D_4 \cdot \bar{R} = 1.864 \times 0.65 = 1.2116$$

$$LCL = D_3 \cdot \bar{R} = 0.136 \times 0.65 = 0.0884$$



All observations are in control.

The process is not under control & it may be due to any one of the assignable reasons like m/c setting change, improper training of operator, defective raw material, tool wear, m/c vibrations etc.

Attribute control chart -

1) p-chart (Proportion or fraction defective chart) -

Sample No.	Sample size	defective	Proportion defective $(p_i = d_i/n_i)$
1	n_1	d_1	$p_1 = d_1/n_1$
2	n_2	d_2	$p_2 = d_2/n_2$
3	n_3	d_3	$p_3 = d_3/n_3$
⋮	⋮	⋮	⋮
N	n_N	d_N	$p_N = d_N/n_N$

Average Proportion defective, $\bar{p} = \dfrac{p_1 + p_2 + p_3 + \cdots + p_N}{N}$

Average sample size, $\bar{n} = \dfrac{n_1 + n_2 + n_3 + \cdots + n_N}{N}$

$CL = \bar{p}$

$UCL = \bar{p} + 3 \cdot \sigma_{\bar{p}} = \bar{p} + 3 \cdot \sqrt{\dfrac{\bar{p}(1-\bar{p})}{\bar{n}}}$

~~$LCL = \bar{p} - 3 \cdot \sigma_{\bar{p}}$~~

$LCL = \bar{p} - 3 \cdot \sigma_{\bar{p}} = \bar{p} - 3 \cdot \sqrt{\dfrac{\bar{p}(1-\bar{p})}{\bar{n}}}$

where, $\sigma_{\bar{p}} = \sqrt{\dfrac{\bar{p}(1-\bar{p})}{\bar{n}}}$

P-charts are used when we can compute the total sample size & the no. of defective. It is prefered for the condition where the sample size is variable.

nP-chart (Number of defective chart) —

$$\boxed{n_1 = n_2 = n_3 = \cdots\cdots = n_N = \bar{n} = n}$$

$$CL = n \cdot \bar{P}$$

$$UCL = n\bar{P} + 3 \cdot \sqrt{n\bar{P}(1-\bar{P})}$$

$$LCL = n\bar{P} - 3 \cdot \sqrt{n\bar{P}(1-\bar{P})}$$

These charts are prefered for the situation where sample size is constant throughout.

2) C-chart (Count of defect chart) —

It is used where we can compute the no. of defects but can not compute the proportion defect. Defects are ~~rad~~ random therefore c-chart is assumed to follow Poission's distribution.

↳ $\boxed{\text{Variance} = \text{Mean}}$

$$\sigma^2 = \bar{c} \Rightarrow \boxed{\sigma = \sqrt{\bar{c}}}$$

$$CL = \bar{c}$$

$$UCL = \bar{c} + 3 \cdot \sqrt{\bar{c}}$$

$$LCL = \bar{c} - 3 \cdot \sqrt{\bar{c}}$$

Acceptance Sampling –

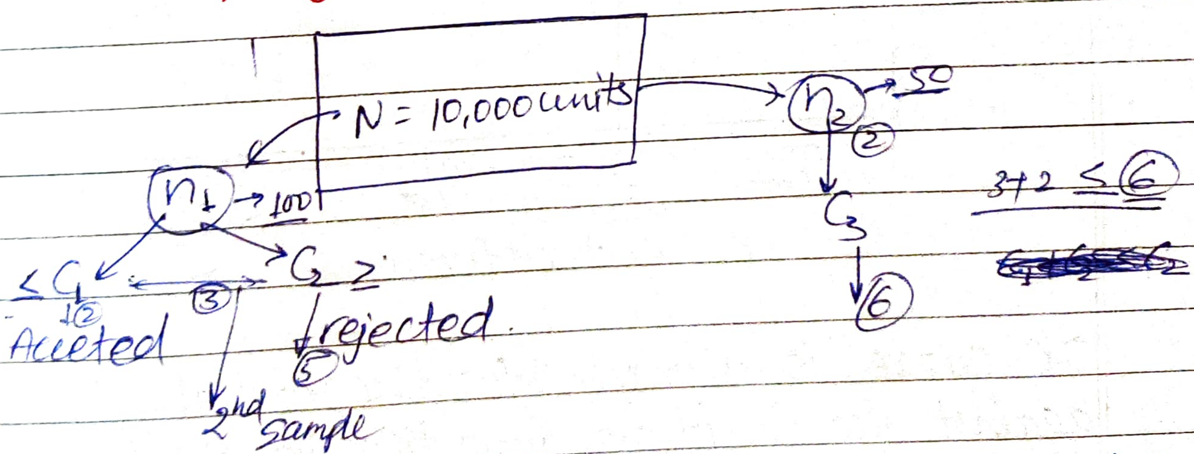
In this method of inspection a sample of goods is randomly inspected in order to decide whether to accept or reject the entire lot. It is used where inspecting every item is either physically not possible & would be very expensive. It is the only method of inspection when testing is done through destructive method.

Sampling Plan –

1) Single Sampling Plan –

Sample is taken once & if the no. of defective are equal to or less than the acceptance no. the entire lot is accepted, otherwise rejected.

2) Double Sampling Plan –

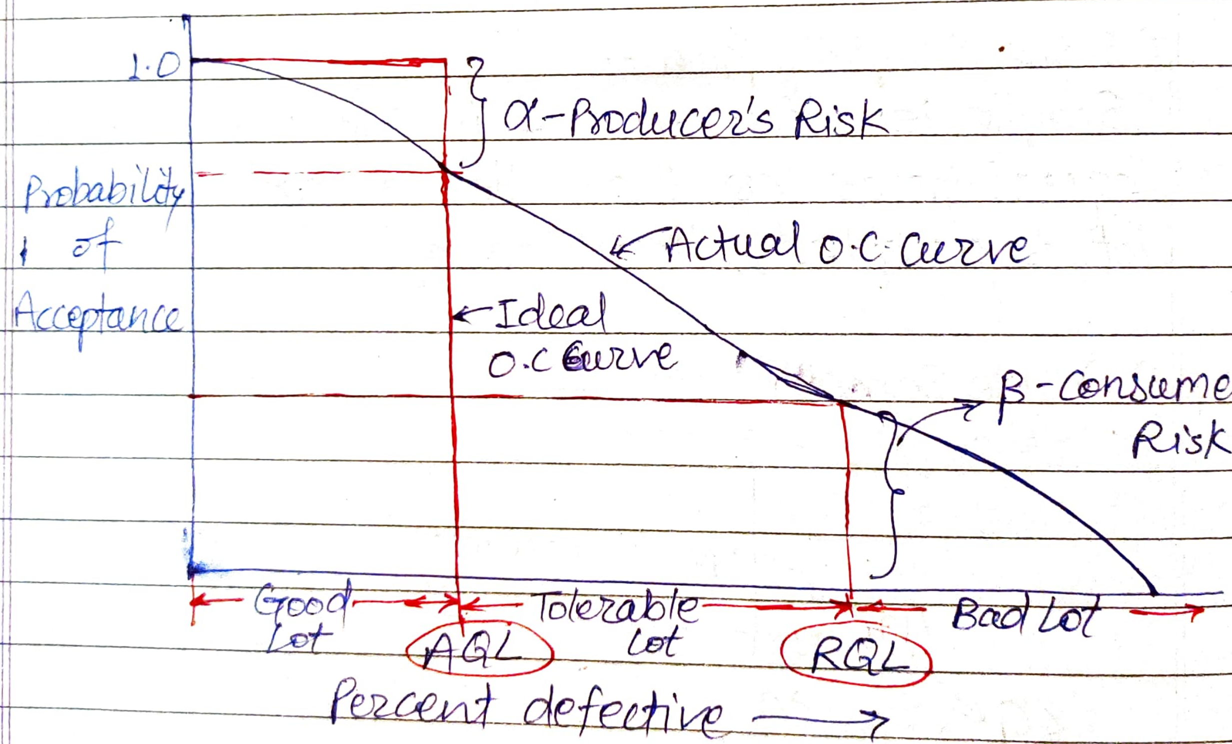


A sample of n_1 units is taken randomly & if the no. of defective are C_1 or less the entire lot is accepted, if it C_2 or more the entire lot is rejected. & if the no. of defective are b/w C_1 & C_2

then another sample of n_2 items is taken randomly. If the total no. of defective from the two samples together is C_3 or less the entire lot is accepted, otherwise rejected.

Note→ As the no. of sampling plan increases average no. of units inspected & chances of making error decreases.

Operating Characteristic Curve:—



O.C. Curve is a graph b/w the probability of acceptance against the fraction defective in a lot. The terms associated with OC-curve are

1) Acceptance Quality level (AQL):— There is a small % defective which consumer's do not have any problem in accepting, AQL representing that Level.

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2) Rejectable Quality level (RQL) –
consumers normally tolerate a few more defective above AQL but then comes a limit beyond which they do not accept any more defective, RQL reprent that level.

3) Producer Risk – (α) –
It represents the probability of rejecting a good quality lot having percent defective equal to AQL (typ-I error)

4) Consumer Risk (β) –
It represents the probability of accepting a bad quality lot having % defective equal to RQL (type-II error).

5) Average Outgoing Quality (AOQ) –
It is a term used to represent the average percent defective in the outgoing product after inspection.

$$\boxed{AOQ = P \cdot Pa \left(\frac{N-n}{N}\right)}$$

if $N >>> n$

$$\boxed{AOQ = P \cdot Pa}$$

where →
$P \rightarrow$ Percent defective
$Pa \rightarrow$ Probability of acceptance
$N \rightarrow$ lot size
$n \rightarrow$ sample size.

metal shaft of dia. ~~25.00 ± 0.020~~ 2.500 ± 0.020 cm. STD, σ estimated approx → 0.005. If sample size is 4, then $\sigma_{\bar{x}} = 0.005/\sqrt{4} = 0.0025$

$UCL_{\bar{x}} = 2.500 + 3(0.0025) = 2.5075$

$LCL_{\bar{x}} = 2.500 - 3(0.0025) = 2.4925$